

THE INTERVIEW

Case Study

AI-Powered Mock Interview & Technical Assessment System

PLATFORM TYPE

Career Preparation & Technical Hiring Simulation

CORE TECHNOLOGY

Google Gemini 2.5-Flash · .NET 10 · Angular v21 · RabbitMQ · Stripe

ARCHITECTURE

Microservices with Event-Driven Saga Pattern

1. Project Overview

THE INTERVIEW is an AI-powered career preparation platform designed to simulate technical interview and assessment workflows. By integrating advanced Generative AI with a robust microservices architecture, the platform provides candidates with dynamic, non-repetitive interview experiences and proctored technical assessments.

2. Problem Statement

Traditional mock interview platforms often rely on static question banks, leading to predictability and a lack of personalisation. Existing tools frequently struggle with:

- **Predictability:** Candidates encounter the same questions repeatedly, reducing genuine preparation value.
- **Surface-Level Evaluation:** Feedback is limited to Correct/Incorrect without deep technical reasoning.
- **Scalability Issues:** Monolithic architectures that become slow and difficult to maintain under load.
- **Integrity Concerns:** Absence of proctoring mechanisms during remote assessments.

3. The Solution

THE INTERVIEW addresses these challenges through a distributed, AI-first approach:

- **Dynamic AI Orchestration:** Utilising Google Gemini 2.5-Flash to generate unique, domain-specific questions in real time.
- **Microservices Architecture:** Decomposing the system into specialised services — Identity, Interview, Assessment, Subscription, and Notification — ensuring independent scalability and fault tolerance.
- **Real-Time Evaluation:** An AI evaluation loop that analyses candidate answers for technical accuracy, providing immediate scores and qualitative feedback.
- **Proctoring:** Browser-level mechanisms including tab-switch detection, fullscreen mode, warnings, and auto-submission on repeated violations.

4. Key Features

- **AI-Driven Interviews:** Tailored sessions for roles such as .NET Developer and Java Engineer, with real-time scoring.
- **Proctored Assessments:** Domain-specific MCQ tests with automated grading and attempt tracking.
- **Premium Tier:** A subscription-based model unlocking unlimited access and detailed performance result breakdowns.
- **Event-Driven Synchronisation:** Using the Saga pattern via RabbitMQ to keep payment and identity states eventually synchronized.
- **User Dashboard:** Comprehensive tracking of progress, grades, and AI-generated feedback.

5. Technical Implementation

The platform is built on a carefully selected technology stack optimised for performance, scalability, and developer productivity:

Layer	Technology	Details
Frontend	Angular v21.2.0	Responsive SPA with RxJS for HTTP flows and client state handling
Backend	.NET Web API v10.0 (net10.0)	Robust, high-performance APIs with EF Core.
AI Integration	Google Gemini API	Utilizing the gemini-2.5-flash model for content generation.
Messaging	RabbitMQ	Handles asynchronous communication and the Subscription Saga.
Payments	Stripe API	Secure, PCI-compliant payment processing.
Gateway	Ocelot	Centralized entry point with JWT validation.

6. Architectural Challenges & Solutions

6.1 AI Latency & Rate Limiting

Challenge: High latency and 429 (Too Many Requests) errors from the LLM provider.

Solution: Migrated to the optimised Gemini 2.5-Flash model and implemented a Polly-based retry policy to handle transient errors gracefully.

6.2 Data Consistency in Distributed Systems

Challenge: Keeping user premium status synchronised across multiple databases after a payment event.

Solution: Implemented the Saga Pattern using RabbitMQ. When a payment is confirmed via a Stripe Webhook, an event is broadcast to update the Identity database asynchronously.

6.3 Assessment Integrity

Challenge: Candidates switching tabs or searching for answers during live assessments.

Solution: Integrated Angular HostListeners to detect tab switching and window focus loss, triggering automatic warnings and eventual auto-submission on violation.

7. Results & Impact

- **Non-Repetitive Learning:** Candidates can take multiple interviews in the same domain without encountering duplicate questions.
- **Actionable Insights:** AI-generated feedback provides specific technical pointers, helping candidates identify weak spots.
- **Seamless Experience:** The Ocelot Gateway ensures that despite the complex microservices topology, the user interacts with a single, unified interface.

8. Conclusion

THE INTERVIEW is an AI-powered candidate preparation platform for mock interviews and technical assessments. By leveraging Generative AI and a resilient microservices architecture, it delivers a premium, scalable, and effective tool for technical interview preparation.

For detailed architecture and API definitions, refer to the High-Level Design (HLD) and Low-Level Design (LLD) documentation.
